think that the OpenStack project took off really quickly. It is not even three years old and we already have 10,000 individual members and 200 companies working on the project. We've had six software releases and expanded the scope from Compute and Object Storage to include projects like Networking, Block Storage, Identity and Dashboard, not to mention a strong ecosystem of open source projects that now surrounds OpenStack. We have users from all around the world and from many different industries , academics and government agencies. I don't know of many open source projects that have moved so quickly.

Q: What are the contributions of these heavyweights to this project?

I have worked with contributors coming from Red Hat, Rackspace, IBM and the likes. What they are contributing is actual code that makes OpenStack work. For every release of OpenStack, we have hundreds of developers that contribute to the software. Most of the developers work for these companies. Each company and contributor brings their own expertise. We have companies like VMware and Red Hat, who are experts in virtualisation and contribute code directly to OpenStack. We have established leaders like NetApp and EMC and innovative startups like SolidFire that contribute to the storage piece. These companies allow us to have experts on different technologies. In addition, they also fund the activities related to promotion and working of the foundation, and allow us to be able to grow the community and support our community worldwide. It is one of the major factors that is contributing to the growth of OpenStack Foundation and OpenStack project. Though we have volunteers associated to the project, having full-time employees contributing to the OpenStack project everyday helps the continuous growth of the project.

Q: How has OpenStack's latest release, Grizzly, been received and what are the key pain points that you have addressed in this release?

The biggest focus of Grizzly release is around usability, scalability and making it easier to operate. In the last release, we had a lot of inputs from the users who have been running OpenStack in production, which we have incorporated. The Grizzly release is a clear indication of the maturity of the OpenStack software development process, as contributors continue to produce a stable, scalable and feature-rich platform for building public, private and hybrid clouds. The community delivered another packed release on schedule, attracting contributions from some of the brightest technologists across virtualization, storage, networking, security, and systems engineering. They are not only solving the complex problems of cloud, but driving the entire technology industry forward. It has nearly 230 new features to support production operations at scale and greater integration with enterprise technologies, including broad Software-Defined Networking support. With more organisations running OpenStack in production, the Grizzly development cycle focused on supporting practical use cases for deployers and operators.

About OpenStack

OpenStack is a cloud computing project that was set up to provide infrastructure as a service (IaaS). It is free open source software released under the terms of the Apache License. The project is managed by the OpenStack Foundation, a non-profit corporate entity established in September 2012 to promote, protect and empower OpenStack software and its community. More than 150 companies joined the project, among which are Intel, AMD, Canonical, SUSE Linux, Inktank, Red Hat, Groupe Bull, Cisco, Dell, Ericsson, HP, IBM, NEC, VMware, Brocade Communications Systems and Yahoo!

The technology consists of a series of interrelated projects that control pools of processing, storage and networking resources throughout a data centre, all managed through a dashboard that gives administrators control while empowering its users to provision resources through a Web interface. The OpenStack community collaborates around a six-month, time-based release cycle with frequent development milestones. During the planning phase of each release, the community gathers for the OpenStack Design Summit to facilitate developer working sessions and assembly plans.

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Grizzly brings additional support for VMware ESX and Microsoft Hyper-V hypervisors. While OpenStack previously had the strongest support for KVM and XEN, this shift helps us reach a broader base of users with the Grizzly release and beyond. With a new 'Cells' capability, Grizzly addressed the scalability issue for very large service providers and private clouds. It allows the users to manage multiple OpenStack compute environments as a single unit. Users are exposed to a single API endpoint and a single control system but there can be a whole nest of clusters underneath.

Grizzly also expands the use cases for block storage. It delivers a full storage service for managing heterogeneous storage environments from a centralised access point. A new intelligent scheduler allows cloud end users to allocate storage based on the workload, whether they are looking for performance, efficiency, or cost effectiveness. The community also added drivers for a diverse selection of backend storage devices, including Ceph/RBD, Coraid, EMC, Hewlett-Packard, Huawei, IBM, NetApp, Red Hat/Gluster, SolidFire and Zadara. END